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United States Patent Application Publication

20250281865

Kind Code

A1

Publication Date

September 11, 2025

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AIR CLEANER

Abstract

Disclosed is an air cleaner capable of preventing an air return phenomenon. The air cleaner includes a case including a suction port disposed on a peripheral surface of the case, a first air cleaning module disposed in the case, the first air cleaning module including a first blower fan, a first filter configured to remove foreign matter from air introduced through the suction port, and a first discharge port disposed at a bottom of the first air cleaning module, the first discharge port being configured to discharge air filtered by the first filter in a downward direction, and a second air cleaning module disposed in the case, the second air cleaning module including a second blower fan, a second filter configured to remove foreign matter from air introduced through the suction port, and a second discharge port disposed at a top of the second air cleaning module.

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Appl. No.: 19/073794

Filed: March 07, 2025

Foreign Application Priority Data

KR

10-2024-0033181

Mar. 08, 2024

Publication Classification

Int. Cl.: B01D46/00 (20220101); B01D46/24 (20060101); B01D46/42 (20060101); B01D46/58 (20220101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Korean Patent Application No. 10-2024-0033181, filed on Mar. 8, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present disclosure relates to an air cleaner, and more particularly to an air cleaner capable of discharging purified air in various directions.

2. Description of the Related Art

[0003] An air cleaner is a device that filters air in a certain space and discharges the filtered air, thereby reducing the concentration of dust or bacteria in the air in the corresponding space. The air cleaner filters out foreign matter by generating a flow of air in the corresponding space, and discharges air with the foreign matter removed therefrom.

[0004] In order to quickly purify air in the indoor space, a discharge port may be formed in an upper portion of an air cleaner, and filtered air discharged upwardly may be guided so as to flow in all directions.

[0005] Further, in the case in which a discharge port is formed in an upper portion of an air cleaner, a separate fan for regulation of the direction of air discharged may be disposed at an upper portion of the air cleaner in order to cause filtered air to flow a long distance.

[0006] Research on a multistage air cleaner in which blowing devices are stacked vertically is underway.

[0007] Related Art Document 1 (Korean Patent Laid-Open Publication No. 10-2022-0083719) and Related Art Document 2 (Korean Patent Laid-Open Publication No. 10-2017-0140578) disclose air cleaners, in each of which blowing devices are disposed in a vertical direction to filter a large amount of air.

[0008] Such an air cleaner of the related art has an advantage of increasing blowing capacity and suctioning and discharging air in a wider variety of directions, thereby easily purifying the surrounding air regardless of whether a user in the room is sitting or standing.

[0009] However, in the air cleaner of the related art, air discharged from the lower blowing device is strongly blown upwardly, thus causing an air return phenomenon. That is, purified air discharged from the lower blowing device may be introduced into the upper blowing device.

[0010] Further, if the amount of air flowing upward is excessively large, the purified air may not circulate over a long distance.

[0011] Further, if the air returning phenomenon occurs, the air flowing back into the upper blowing device acts as an obstacle that impedes introduction of impure outside air into the upper blowing device, thus reducing the flow amount of air in the upper blowing device.

[0012] In order to solve the above-described problems, each of the air cleaners of the related art disclosed in Related Art Document 1 (Korean Patent Laid-Open Publication No. 10-2022-0083719) and Related Art Document 2 (Korean Patent Laid-Open Publication No. 10-2017-0140578) includes a partition plate disposed between the plurality of blowing devices in order to change the flow direction of air, thereby preventing the air return phenomenon.

[0013] However, because the flow amount of air that collides with the partition wall and flows in the lateral direction is smaller than the flow amount of air rising in the radially outward direction, it is not possible to sufficiently change the flow direction of the rising air discharged from the lower blowing device, and thus the discharged air may rise in the substantially oblique direction and may flow back into the upper blowing device.

[0014] If the height at which the partition plate is located is lowered in order to increase the flow amount of air that collides with the partition wall and flows in the lateral direction, an air discharge area may be reduced.

[0015] If the air cleaner employs an airflow guide structure for preventing the air return phenomenon, such as a partition plate, the overall height and size of the product may increase, and the exterior of the product may appear discontinuous.

[0016] Further, in the air cleaners disclosed in the above related art documents, the peripheral grille is divided into an upper part and a lower part, which causes increase in the number of fastening portions and inconvenience in management and cleaning of components. Furthermore, in the air cleaners disclosed in the above related art documents, because the suction port is blocked by the pillar of the filter mounting part, flow resistance may occur, and thus air cleaning performance may be reduced.

SUMMARY OF THE INVENTION

[0017] An aspect of the present disclosure is directed to providing an air cleaner capable of preventing discharged purified air from flowing back thereinto, thereby improving air cleaning performance.

[0018] Another aspect of the present disclosure is directed to providing an air cleaner capable of preventing an air return phenomenon without adding a structure such as a partition plate, thereby having a reduced size.

[0019] Still another aspect of the present disclosure is directed to providing an air cleaner capable of forming an airflow that effectively removes allergens causing allergies. Still another aspect of the present disclosure is directed to providing an air cleaner capable of improving suction capacity of air suctioned into the air cleaner.

[0020] Still another aspect of the present disclosure is directed to providing an air cleaner capable of facilitating coupling and separation of a case forming the external appearance of the air cleaner.

[0021] Still another aspect of the present disclosure is directed to providing an air cleaner capable of simplifying a case coupling structure and reducing manufacturing costs.

[0022] Still another aspect of the present disclosure is directed to providing an air cleaner enabling convenient replacement of a filter.

[0023] Still another aspect of the present disclosure is directed to providing an air cleaner capable of facilitating maintenance and repair of internal parts.

[0024] Still another aspect of the present disclosure is directed to providing an air cleaner capable of causing discharged purified air to circulate over a longer distance.

[0025] The aspects of the present disclosure are not limited to the aspects mentioned above, and other aspects not mentioned herein will be clearly understood by those skilled in the art from the following description.

[0026] An air cleaner according to an embodiment of the present disclosure discharges purified air through the bottom of the product, thereby preventing the discharged air from being introduced thereinto through a suction port formed in an upper side of the product.

[0027] An air cleaner according to an embodiment of the present disclosure is configured such that an air cleaning module configured to discharge purified air through the top thereof and an air cleaning module configured to discharge purified air through the bottom thereof are stacked vertically, thereby preventing an air return phenomenon and improving an air cleaning performance.

[0028] An air cleaner according to an embodiment of the present disclosure does not require a

structure for preventing the air return phenomenon, thereby having a reduced size and improving freedom of design, such as placement of internal parts.

[0029] An air cleaner according to an embodiment of the present disclosure is configured to form an airflow at a height at which allergens causing allergies are mainly located, thereby effectively removing the allergens.

[0030] An air cleaner according to an embodiment of the present disclosure suctions air through a suction port formed in the peripheral surface of a case, thereby improving suction performance.

[0031] An air cleaner according to an embodiment of the present disclosure exhibits improved performance of suctioning air into a filter while securing sufficient support rigidity through a truss structure.

[0032] An air cleaner according to an embodiment of the present disclosure is configured such that a single case covers the outer circumferential surfaces of all of air cleaning modules stacked vertically, rather than being divided into a plurality of parts for covering the respective air cleaning modules, thereby facilitating coupling and separation of the case and reducing manufacturing costs.

[0033] An air cleaner according to an embodiment of the present disclosure includes a case having a suction port formed in the peripheral surface thereof and an air cleaning module including a blower fan disposed in the case and a filter configured to remove foreign matter from air introduced through the suction port, the air cleaning module includes a first discharge port formed in the bottom thereof and a second discharge port formed in the top thereof, and air with foreign matter removed therefrom is discharged downwardly through the first discharge port and is discharged upwardly through the second discharge port, thereby preventing an air return phenomenon.

[0034] The air cleaning module includes a first air cleaning module and a second air cleaning module disposed above the first air cleaning module.

[0035] The first discharge port is formed in the bottom of the first air cleaning module, and the second discharge port is formed in the top of the second air cleaning module.

[0036] The air cleaner may further include a booster module disposed above the air cleaning module to regulate the direction of air discharged through the second discharge port, thereby sending purified air farther away.

[0037] The air cleaner may further include a bottom plate disposed below the air cleaning module so as to be in contact with the floor.

[0038] An airflow guide may be disposed on an upper surface of the bottom plate so as to guide air discharged downwardly through the first discharge port in a lateral direction.

[0039] The airflow guide may include a curved portion, a flat portion extending laterally from the curved portion, and a boss protruding upward from an end portion of the flat portion.

[0040] A lower discharge grille including a plurality of grilles bent in an outward direction may be disposed inside the first discharge port.

[0041] The booster module may include an outer cover, an inner grille disposed inside the outer cover and having a discharge port formed therein, a rear suction grille having a suction port formed therein, a booster fan disposed on a front surface of the rear suction grille and disposed inside the inner grille, and a motor configured to rotate the booster fan.

[0042] The booster module may further include a display panel disposed on a front surface of the booster fan and an inner flow guide disposed between the outer cover and the inner grille to change the flow direction of air discharged through the discharge port of the inner grille.

[0043] The air discharged through the discharge port of the inner grille may pass through a slit formed between the inner flow guide and the display panel.

[0044] The display panel may cover the entirety of the front surface of the booster fan.

[0045] The motor may be disposed inside the booster fan.

[0046] The booster module may be removably mounted to the top of the second air cleaning module.

[0047] The booster module may further include a power module configured to come into contact

with a terminal of the second air cleaning module and receive power from the second air cleaning module when the booster module is mounted to the second air cleaning module.

[0048] The case may include a first case having a semi-cylindrical shape and covering a portion of the outer circumferential surface of the air cleaning module and a second case having a semi-cylindrical shape and covering the remaining portion of the outer circumferential surface of the air cleaning module.

[0049] The air cleaning module may include a first magnet and a second magnet disposed above the first magnet.

[0050] The first case may include a first lower metallic piece coupled to the first magnet and a first upper metallic piece coupled to the second magnet.

[0051] The second case may include a second lower metallic piece coupled to the first magnet and a second upper metallic piece coupled to the second magnet.

[0052] The air cleaning module may include a fan housing accommodating the blower fan and a plurality of support parts extending vertically from the fan housing.

[0053] Some of the plurality of support parts may be formed in a truss structure in which a plurality of linear members is arranged in a plurality of triangular forms.

[0054] The details of other embodiments will be included in the detailed description and the drawings below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] The above and other objects, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0056] FIG. 1 is a perspective view of an air cleaner according to an embodiment of the present disclosure;

[0057] FIG. 2 is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure;

[0058] FIG. 3 is a view showing an airflow generated by the air cleaner according to the embodiment of the present disclosure;

[0059] FIG. 4 is an exploded view of the air cleaner according to the embodiment of the present disclosure;

[0060] FIG. 5 is an assembled view of a booster module according to the embodiment of the present disclosure;

[0061] FIGS. 6 and 7 are exploded views of the booster module according to the embodiment of the present disclosure;

[0062] FIG. 8 is a cross-sectional view of the booster module according to the embodiment of the present disclosure;

[0063] FIG. 9 is a view showing an airflow in the booster module according to the embodiment of the present disclosure;

[0064] FIG. 10 is an assembled view of a first air cleaning module according to the embodiment of the present disclosure;

[0065] FIGS. 11 and 12 are exploded views of the first air cleaning module according to the embodiment of the present disclosure;

[0066] FIG. 13 is an assembled view of a control module according to the embodiment of the present disclosure;

[0067] FIGS. 14 and 15 are exploded views of the control module according to the embodiment of the present disclosure;

[0068] FIG. **16** is an assembled view of a second air cleaning module according to the embodiment of the present disclosure;

[0069] FIGS. **17** and **18** are exploded views of the second air cleaning module according to the embodiment of the present disclosure;

[0070] FIGS. **19** to **21** are views for explaining discharge of purified air through the bottom of the air cleaner according to the embodiment of the present disclosure;

[0071] FIGS. **22A** to **22E** are views for explaining performance comparison between a case in which purified air is discharged through the upper and lower ends of the product and a case in which purified air is discharged only through the upper end of the product;

[0072] FIGS. **23** and **24** are views for explaining a case coupling structure according to the embodiment of the present disclosure; and

[0073] FIGS. **25** to **28** are views for explaining the structure of support parts according to the embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0074] Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. The present disclosure may, however, be embodied in many different forms, and should not be construed as being limited to the embodiments set forth herein.

[0075] In the drawings, illustration of parts unrelated to the description is omitted to clearly and briefly describe the present disclosure, and the same or extremely similar components are denoted by the same reference numerals throughout the specification.

[0076] As used herein, the terms with which the names of components are suffixed, “module” and “unit”, are assigned to facilitate preparation of this specification, and are not intended to suggest unique meanings or functions. Accordingly, the terms “module” and “unit” may be used interchangeably.

[0077] It will be understood that although the terms “first”, “second”, etc., may be used herein to describe various components, these components should not be limited by these terms. These terms are only used to distinguish one component from another component.

[0078] FIG. **1** is a perspective view of an air cleaner according to an embodiment of the present disclosure. FIG. **2** is a cross-sectional view of the air cleaner according to the embodiment of the present disclosure, and FIG. **3** is a view showing an airflow generated by the air cleaner according to the embodiment of the present disclosure. FIG. **4** is an exploded view of the air cleaner according to the embodiment of the present disclosure.

[0079] Referring to FIGS. **1** to **4**, the air cleaner **1** includes a case **10** forming the external appearance thereof. The case **10** may have suction ports **11a** and **12a** formed in the peripheral surface thereof to suction air from various directions. Air may be suctioned from all directions with respect to a central line passing through the internal center of the case **10** in an upward-downward direction. The suction ports **11a** and **12a** are provided in plural, and the plurality of suction ports **11a** and **12a** is spaced apart from each other in the circumferential direction of the case **10**. The plurality of suction ports **11a** and **12a** is evenly formed in the circumferential direction along the peripheral surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0080] In this specification, the upward-downward direction or the vertical direction is defined as an axial direction. The axial direction may correspond to the central axis-direction of blower fans **140** and **240**, which will be described later, i.e., the motor shaft direction of the fans. The radial direction or the horizontal direction may be understood as a direction perpendicular to the axial direction. The circumferential direction may be understood as a circumferential direction of an imaginary circle that is formed when rotating about the axial direction with a distance in the radial direction as a rotational radius.

[0081] The air cleaner **1** includes air cleaning modules **100** and **200** that generate airflows. The air

cleaning modules **100** and **200** include blower fans **140** and **240**, which are disposed in the case **10**, and filters **120** and **220**, which remove foreign matter from air introduced through the suction ports **11a** and **12a**, respectively.

[0082] The air introduced through the suction ports **11a** and **12a** may pass through the filters **120** and **220**. Each of the filters **120** and **220** may be formed in a cylindrical shape and may have a filter surface that filters air.

[0083] The case **10** may have a cylindrical shape. The case **10** may include a first case **11**, which has a semi-cylindrical shape and covers portions of the outer circumferential surfaces of the air cleaning modules **100** and **200**, and a second case **12**, which has a semi-cylindrical shape and covers the remaining portions of the outer circumferential surfaces of the air cleaning modules **100** and **200**.

[0084] Alternatively, the case **10** may have a shape of a truncated cone (a cone with the top cut off), and each of the first case **11** and the second case **12** may have a shape of a semi-truncated cone.

[0085] The first case **11** and the second case **12** may be coupled to each other to form the external appearance of the air cleaner **1**. Because the first case **11** covers the front surface of the air cleaner **1**, the first case **11** may be referred to as a “front case” or a “front cover”. Because the second case **12** covers the rear surface of the air cleaner **1**, the second case **12** may be referred to as a “rear case” or a “rear cover”.

[0086] The suction ports **11a** and **12a** include a first suction port **11a** formed in the first case **11** and a second suction port **12a** formed in the second case **12**. The suction ports **11a** and **12a** allow the inside of the case **10** and the outside to communicate with each other. The suction ports **11a** and **12a** are provided in plural. The plurality of suction ports **11a** and **12a** may be formed in the peripheral surface of the case **10**.

[0087] The plurality of suction ports **11a** and **12a** is evenly formed in the circumferential direction along the outer circumferential surface of the case **10** so that air is capable of being suctioned into the case **10** from any direction.

[0088] The plurality of suction ports **11a** and **12a** is formed in a stripe shape that is elongated in the upward-downward direction. Alternatively, the plurality of suction ports **11a** and **12a** may be formed through perforation in a circular or elliptical shape.

[0089] As described above, because the case **10** is formed in a cylindrical shape and the plurality of suction ports **11a** and **12a** is formed along the outer circumferential surface of the case **10**, the amount of air suctioned may increase.

[0090] Referring to FIGS. **1** to **4**, the air cleaning modules **100** and **200** include a first air cleaning module **100** and a second air cleaning module **200** that are disposed in the upward-downward direction.

[0091] For example, the second air cleaning module **200** may be disposed above the first air cleaning module **100**. Because the first air cleaning module **100** is disposed at a lower portion of the air cleaner **1**, the first air cleaning module **100** may be referred to as a “lower air cleaning module” or a “lower module”, and because the second air cleaning module **200** is disposed at an upper portion of the air cleaner **1**, the second air cleaning module **200** may be referred to as an “upper air cleaning module” or an “upper module”.

[0092] The first air cleaning module **100** and the second air cleaning module **200** include blower fans **140** and **240** and filters **120** and **220** configured to remove foreign matter from air introduced through the suction ports **11a** and **12a**, respectively.

[0093] The first air cleaning module **100** includes a first blower fan **140** and a first filter **120** configured to remove foreign matter from air introduced through the suction ports. In addition, the first air cleaning module **100** further includes a first fan motor **130** configured to rotate the first blower fan **140**.

[0094] The second air cleaning module **200** includes a second blower fan **240** and a second filter **220** configured to remove foreign matter from air introduced through the suction ports. In addition,

the second air cleaning module **200** further includes a second fan motor **230** configured to rotate the second blower fan **240**.

[0095] The air cleaning modules **100** and **200** include a first discharge port **110a** formed in the bottom of the air cleaning module **100** and a second discharge port **210a** formed in the top of the air cleaning module **200**. The first discharge port **110a** is formed below the first blower fan **140** and the first filter **120**. The second discharge port **210a** is formed above the second blower fan **240** and the second filter **220**. Air with foreign matter removed therefrom by the filters **120** and **220** is discharged downwardly through the first discharge port **110a** and is discharged upwardly through the second discharge port **210a**.

[0096] In the case in which the second air cleaning module **200** is disposed above the first air cleaning module **100**, the first discharge port **110a** is formed in the bottom of the first air cleaning module **100**, and the second discharge port **210a** is formed in the top of the second air cleaning module **200**.

[0097] In addition, a lower discharge grille **110** including a plurality of grilles is disposed at the bottom of the first air cleaning module **100** in order to guide a lower airflow. In addition, an upper discharge grille **210** including a plurality of grilles is disposed at the top of the second air cleaning module **200** in order to guide an upper airflow.

[0098] The upper discharge grille **210** may include an inner wall **211** and an outer wall **212**, which have a cylindrical shape. The plurality of grilles of the upper discharge grille **210** may interconnect the inner wall **211** and the outer wall **212** and may be disposed in a radial form.

[0099] The lower discharge grille **110** may include a plurality of grilles having a concentric circle structure. The plurality of grilles may be a plurality of circles having the same center and different radii. Each of the plurality of grilles may include a section bent in the radially outward direction. The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0100] A bottom plate **170** is disposed below the first air cleaning module **100**. The bottom plate **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0101] An airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction. The bottom plate **170** may further include a base **170b** that supports the airflow guide **170a**. The airflow guide **170a** may be formed on the upper surface of the base **170b**. In addition, the airflow guide **170a** and the base **170b** may be coupled to each other to constitute the bottom plate **170**.

[0102] Air with foreign matter removed therefrom by the first filter **120** is discharged in the downward direction through the first discharge port **110a**. Air with foreign matter removed therefrom by the second filter **220** is discharged in the upward direction through the second discharge port **210a**. According to the present disclosure, although two air cleaning modules are stacked vertically, air is not discharged upwardly from the lower air cleaning module. Therefore, while suction performance is improved by forming the suction ports **11a** and **12a** in the peripheral surface of the case **10**, it is possible to prevent air discharged from the lower air cleaning module from flowing back into the upper air cleaning module.

[0103] The 2-stage air cleaner of the related art has a problem in that purified air discharged from the lower air cleaning module flows upward and is suctioned into the upper air cleaning module, whereby air cleaning performance is reduced. In contrast, according to the present disclosure, purified air discharged from the lower air cleaning module flows downward, and thus introduction of the purified air into the upper air cleaning module is prevented, whereby air cleaning performance is improved.

[0104] Further, according to the present disclosure, since the air cleaner **1** does not need to have a structure for preventing introduction of purified air discharged therefrom or a configuration for discharge, the overall size of the product may be further reduced, and the freedom of design such as

placement of internal parts may be improved.

[0105] For example, a control module **400** may be disposed at the center of the air cleaner **1**, thereby reducing the lengths of power lines and signal lines, simplifying a line connection structure, and reducing the overall height of the product. Further, the control module **400** may be disposed between the air cleaning modules **100** and **200** so that the air cleaning modules **100** and **200** are controlled by the single integrated control module **400**. Accordingly, each of the air cleaning modules **100** and **200** does not need to include an individual control module, and thus the overall height of the product may be reduced.

[0106] General allergen particles have a size of about 3 μm to about 100 μm , which is greater than the size of fine dust, and thus tend to settle on the floor. The lower airflow discharged downwardly from the first air cleaning module **100** causes allergen particles to float in the air and thus to be collected by the filters **120** and **220**. In this way, it is possible to effectively remove allergens causing allergies using the lower airflow discharged downwardly from the first air cleaning module **100**.

[0107] In addition, the air cleaner **1** may include UVC LEDs **481** and **482** (refer to FIG. **14**), which are disposed in the case **10** and output ultraviolet in a wavelength band that has a sterilization effect on bacteria, mold, and microorganisms, and an ionizer, which is disposed in the case **10** and eliminates germs and mold using electricity supplied thereto.

[0108] In addition, the air cleaning modules **100** and **200** include fan housings **145** and **245** accommodating the blower fans **140** and **240** and a plurality of support parts **150** and **250** extending in the vertical direction from the fan housings **145** and **245**. The plurality of support parts **150** and **250** may extend in the longitudinal direction of the air cleaner **1**. The number of support parts **150** and **250** may be varied depending on the size or specifications of the product. Some of the support parts **150** and **250** may be different in at least one of shape, size, or material from the remaining ones of the support parts **150** and **250**.

[0109] Each of wire covers **155** and **255** may be disposed on at least one surface of a respective one of the plurality of support parts **150** and **250** so as to cover a respective one of the support parts **150** and **250**. The support parts **150** and **250** and the wire covers **155** and **255** may be spaced apart from each other to define spaces in which wires or the like are disposed.

[0110] In addition, the air cleaning modules **100** and **200** may further include wire covers **115** and **215**, inside which wires or the like are disposed. The fan motors **130** and **230** may be accommodated in motor fastening parts **125** and **225**, respectively.

[0111] The air cleaning modules **100** and **200** include filter mounting parts **160** and **260** in which the filters **120** and **220** are mounted, respectively. The control module **400** is disposed between the filter mounting parts **160** and **260**.

[0112] Steel nets **165** and **265** are disposed between the filters **120** and **220** and the blower fans **140** and **240** in order to prevent body parts of the user or any other object from entering the spaces in which the blower fans **140** and **240** are disposed.

[0113] A top cover **270** is disposed on the top of the case **10**. The upper discharge grille **210** is disposed inside the top cover **270**.

[0114] In addition, a booster module **300** that regulates the direction of air discharged through the second discharge port **210a** may be disposed above the air cleaning modules **100** and **200**. The booster module **300** may be disposed above the second air cleaning module **200** in order to change the discharge direction of the airflow generated by the second air cleaning module **200**.

[0115] The booster module **300** may send air discharged from the second air cleaning module **200** farther away. In addition, the booster module **300** may blend an airflow while rotating left and right, thereby forming various types of airflows. In some embodiments, the booster module **300** may move in the upward-downward direction so as to form a wider variety of airflows.

[0116] The booster module **300** may be removably mounted to the top of the second air cleaning module **200**. In addition, a contact power supply structure may be applied to the booster module

300 and the second air cleaning module **200**, so that power is supplied from the second air cleaning module **200** to the booster module **300** when the booster module **300** is mounted to the second air cleaning module **200**.

[0117] When internal parts of the booster module **300**, such as a booster fan **330**, need to be washed, or when the upper discharge grille **210** disposed at the top of the second air cleaning module **200** needs to be cleaned, the booster module **300** may be removed. In this way, it is possible to conveniently keep the product hygienic.

[0118] The booster module **300** includes an outer cover **320**, an inner grille **355** disposed inside the outer cover **320** and having a discharge port **355a** (refer to FIG. 5) formed therein, a rear suction grille **311** having a suction port **311a** (refer to FIG. 6) formed therein, a booster fan **330** disposed on the front surface of the rear suction grille **311** and disposed inside the inner grille **355**, and a motor **340** configured to rotate the booster fan **330**.

[0119] The rear suction grille **311** may be a part of a rear cover **310** that forms the external appearance of the rear surface of the booster module **300**. Alternatively, the rear suction grille **311** may be provided separately and coupled to the rear cover **310**.

[0120] The rear cover **310** may have an opening formed therein so as to allow a portion of the upper end of a booster neck **360** to pass therethrough. In addition, the rear cover **310** may be coupled to the booster neck **360** by means of predetermined fastening member.

[0121] The booster module **300** may be supported by the booster neck **360** and may be mounted to the second air cleaning module **200**.

[0122] The booster module **300** may be provided so as to be movable. The booster module **300** may be maintained in an inclined state or a lying state, as shown in FIG. 1.

[0123] The booster module **300** further includes a display panel **390** disposed on the front surface of the booster fan **330**. The display panel **390** may cover the entirety of the front surface of the booster fan **330**. The motor **340** may be disposed inside the booster fan **330**.

[0124] The booster fan **330** may be a sirocco fan. If the booster fan **330** is implemented as a mixed-flow fan, an airflow is discharged through the front grille. However, the sirocco fan may discharge an airflow through the inner grille **355** formed laterally. Accordingly, it is possible to place the display panel **390** on the front surface of the fan, thereby shielding the fan structure. In addition, the sirocco fan has an advantage of reducing noise compared to the mixed-flow fan.

[0125] In addition, the booster module **300** further includes an inner flow guide **350** disposed between the outer cover **320** and the inner grille **355** in order to change the flow direction of air discharged through the discharge port **355a** of the inner grille **355**.

[0126] A slit **351** is formed between the inner flow guide **350** and the display panel **390**. The air discharged through the discharge port **355a** of the inner grille **355** passes through the slit **351**, and then is discharged from the booster module **300**.

[0127] FIG. 5 is an assembled view of the booster module according to the embodiment of the present disclosure, and FIGS. 6 and 7 are exploded views of the booster module according to the embodiment of the present disclosure.

[0128] FIG. 8 is a cross-sectional view of the booster module according to the embodiment of the present disclosure, and FIG. 9 is a view showing an airflow in the booster module according to the embodiment of the present disclosure.

[0129] Referring to FIGS. 5 to 9, the display panel **390** that displays information about operation of the air cleaner **1** is disposed at the front of the booster module **300**. The display panel **390** may move together with the booster module **300**. The user may more easily view the information displayed on the display panel **390** disposed at the top of the product.

[0130] The booster fan **330** and the booster motor **340** are disposed on the rear surface of the display panel **390**. The booster motor **340** is disposed inside the booster fan **330**. The display panel **390** covers the entireties of the front surfaces of the booster fan **330** and the booster motor **340**, thereby preventing exposure of the booster fan **330** to the user.

[0131] More preferably, the booster fan **330** may be a sirocco fan. If the booster fan **330** is implemented as a mixed-flow fan, air flows forward. Thus, the mixed-flow fan is not suitable for the slit-type flow path, and a front grille is required. However, if the booster fan **330** is implemented as a sirocco fan, resistance to an airflow discharged through the slit structure is reduced. Thus, the sirocco fan is suitable for the slit structure, and a front grille is eliminated. Accordingly, the fan and the inner grille are not visible from the outside.

[0132] The rear cover **310** and the outer cover **320** are coupled to each other to define a space in which parts are accommodated. The booster fan **330** and the booster motor **340** may be disposed in the inner space of the outer cover **320**.

[0133] The inner grille **355** in which the discharge port **355a** is formed and the inner flow guide **350** that changes the flow direction of air discharged through the discharge port **355a** of the inner grille **355** are disposed between the outer cover **320** and the booster fan **330**.

[0134] The inner grille **355** is disposed on the lateral side of the booster fan **330**. The airflow caused by rotation of the booster fan **330** occurs along the discharge port **355a** of the inner grille **355** formed laterally and the inner flow guide **350**.

[0135] An inner flow path **352** including a forwardly curved surface is disposed on the inner surface of the inner flow guide **350**. An end portion of the inner flow guide **350** and an end portion of the display panel **390** may be spaced a predetermined interval from each other to form the slit **351**.

[0136] The air discharged through the discharge port **355a** of the inner grille **355** passes through the inner flow path **352** and the slit **351**, and then is discharged from the booster module **300**.

[0137] According to the present disclosure, because an airflow is discharged through the slit **351**, it is possible to minimize accumulation of dust on the booster fan **330** and the grille **355**. Further, because the fan **330** and the grille **355** are not visible from the outside, the aesthetics of the product may be improved.

[0138] According to the present disclosure, an airflow is delivered through the inner grille **355** disposed on the lateral side of the booster fan **330**, the inner flow path **352**, and the slit **351**, the blades of the booster fan **330** are not visible from the outside, and the size of the motor **340** is reduced. Further, according to the present disclosure, because a front grille is eliminated, it is possible to solve a problem of dust accumulating on a front grille.

[0139] The rear cover **310** includes a coupling portion **312** coupled to the outer cover **320** and a grille portion **311** in which the suction port **311a** is formed. The grille portion **311** may include a plurality of grilles, and may be the aforementioned rear suction grille **311**. The rear surface of the outer cover **320** may be formed corresponding to the shape of the coupling portion **312** and the coupling structure. For example, a stepped portion may be formed at the rear side of the outer cover **320**, and a coupling member formed to be coupled to the coupling portion **312** may be disposed on the stepped portion so that the contours of the rear cover **310** and the outer cover **320** are smoothly connected when the rear cover **310** and the outer cover **320** are coupled to each other. The outer cover **320** may be coupled to the inner flow guide **350** using a coupling member formed on the inner surface thereof.

[0140] An assembly of components from the display panel **390** disposed at the front side to the rear suction grille **311** disposed at the rear side, which are disposed at the top of the air cleaner **1**, may be referred to as a “booster head”.

[0141] The booster module **300** (or the booster head) is coupled to and supported by the booster neck **360**. The booster neck **360** is coupled to the air cleaning modules **100** and **200**, which correspond to the main body of the air cleaner **1**.

[0142] The booster neck **360** may be coupled to the top of the second air cleaning module **200**. Accordingly, at least a portion of the purified air discharged through the second discharge port **210a** formed in the top of the second air cleaning module **200** may flow into the suction portion **311a** of the rear suction grille **311**.

[0143] The upper portion of the booster neck **360** may be coupled to the booster module **300**. The lower portion of the booster neck **360** may be connected to the second air cleaning module **200**. In order to further facilitate connection between the booster neck **360** and the second air cleaning module **200**, a protruding portion may be formed at the lower portion of the booster neck **360** so as to protrude downward.

[0144] A head rotating part **380** may be located below the booster neck **360**. The head rotating part **380** may be located inside the inner wall **211** of the upper discharge grille **210**. The head rotating part **380** may include a structure for rotating the booster module **300** and a case for accommodating internal components, and may be connected to or accommodate components such as a stepper motor, an ionizer, and wires.

[0145] The upper end of the head rotating part **380** may be coupled to the booster neck **360**. The lower end of the head rotating part **380** may be coupled to another mechanical structure of the second air cleaning module **200**. For example, the lower end of the head rotating part **380** may be coupled to an upper side of the second fan housing **245**.

[0146] Alternatively, the lower end of the head rotating part **380** may be coupled to an upper side of the second motor fastening part **225**. In another embodiment, a separate coupling part (not shown) may be provided between the lower end of the head rotating part **380** and the upper end of the second fan housing **245** or the second motor fastening part **225**.

[0147] The booster module **300** includes a power module **370**. The power module **370** may include a power line for supply of power, a signal line for transmission of signals, and a terminal connected to the power line and the signal line.

[0148] When the booster module **300** is mounted to the second air cleaning module **200**, the terminal of the power module **370** may come into contact with the terminal of the second air cleaning module **200**, and accordingly, the power module **370** may receive power from the second air cleaning module **200**.

[0149] FIG. **10** is an assembled view of the first air cleaning module according to the embodiment of the present disclosure, and FIGS. **11** and **12** are exploded views of the first air cleaning module according to the embodiment of the present disclosure.

[0150] Referring to FIGS. **10** to **12**, the first discharge port **110a** that is open downward is formed in a lower portion of the first air cleaning module **100**. The lower discharge grille **110** that changes the direction of air flowing downward is disposed inside the first discharge port **110a**.

[0151] The annular first discharge port **110a** is formed in a lower portion of the first air cleaning module **100**. Air with foreign matter removed therefrom by the first filter **120** is discharged in the downward direction through the first discharge port **110a**.

[0152] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0153] The first air cleaning module **100** includes a first filter mounting part **160** to which the first filter **120** is mounted. The first filter **120** may be removably mounted to the first filter mounting part **160**. The first filter **120** may have a hollow cylindrical shape, and air may be introduced through the outer circumferential surface of the first filter **120**. While the air passes through the first filter **120**, impurities such as fine dust may be removed from the air.

[0154] Because the first filter **120** has a cylindrical shape, air is capable of being introduced into the first filter **120** from any direction. Therefore, the air filtering area may increase.

[0155] The first filter mounting part **160** may be formed in a cylindrical shape corresponding to the shape of the first filter **120**. In the process of mounting the first filter **120**, the first filter **120** may be slidably fitted into the first filter mounting part **160**. Conversely, in the process of demounting the first filter **120**, the first filter **120** may be slidably withdrawn from the first filter mounting part **160**.

[0156] The first air cleaning module **100** includes a first fan housing **145** disposed below the first filter **120**, a first blower fan **140** rotatably disposed in the first fan housing **145**, and a first fan

motor **130** configured to rotate the first blower fan **140**. In addition, the first air cleaning module **100** may further include a wire cover **115** that protects a wire connected to the first fan motor **130**. [0157] The first blower fan **140** suctions air in the axial direction and discharges the air downwardly in the radial direction. The first fan housing **145** may form a fan path that accommodates the first blower fan **140** and guides the air moved by the first blower fan **140** downwardly.

[0158] The first fan motor **130** may be supported by the first motor fastening part **125**. The rotation shaft of the first fan motor **130** may extend upward from the first fan motor **130** and may pass through the bottom of the first motor fastening part **125** to be connected to the first blower fan **140**.

[0159] First magnets **180** are disposed on two opposite sides of the first fan housing **145**. The first magnets **180** may be coupled to metallic pieces of the first case **11**. In addition, the first magnets **180** may be coupled to metallic pieces of the second case **12**.

[0160] In some embodiments, each of the first magnets **180** may include a plurality of magnets. Some of the plurality of magnets may be coupled to the metallic pieces of the first case **11**, and the remaining ones of the plurality of magnets may be coupled to the metallic pieces of the second case **12**.

[0161] The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the first fan housing **145**, and the magnets may be disposed at the first case **11** and the second case **12**.

[0162] A plurality of support portions **150** extends upward from the first fan housing **145**. For example, the support portions **150** may be beams elongated in the longitudinal direction. A wire cover **155** may be disposed on at least one surface of each of the plurality of support portions **150** so as to cover the support portions **150**. The support portions **150** and the wire cover **155** may be spaced a predetermined interval from each other to define a space in which wires or the like are disposed.

[0163] A steel net **165** is disposed between the first filter **120** and the first blower fan **140** in order to prevent body parts of the user or any other object from entering the space in which the first blower fan **140** is disposed.

[0164] The bottom plate **170** is disposed below the first air cleaning module **100**. The bottom plate **170** is disposed in contact with the floor to support the air cleaning modules **100** and **200**.

[0165] The airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction. The bottom plate **170** may further include a base **170b** that supports the airflow guide **170a**.

[0166] FIG. **13** is an assembled view of the control module according to the embodiment of the present disclosure, and FIGS. **14** and **15** are exploded views of the control module according to the embodiment of the present disclosure.

[0167] Referring to FIGS. **13** to **15**, the control module **400** includes an upper cover **410** and a lower cover **420** that are coupled to each other to form the external appearance of the control module **400**. The control module **400** includes a main PCB **430** on which circuits, such as a controller for control of overall operation of the air cleaner **1**, are mounted, and the main PCB **430** is accommodated in an inner space defined by the upper cover **410** and the lower cover **420**.

[0168] Each of the upper cover **410** and the lower cover **420** has a central cavity formed therein, and an upper filter pressing plate **440** and a lower filter pressing plate **450** are disposed in the central cavity of the upper cover **410** and the central cavity of the lower cover **420**, respectively. Each of the upper filter pressing plate **440** and the lower filter pressing plate **450** may be connected to a plurality of springs **435**.

[0169] The control module **400** may be located between the first air cleaning module **100** and the second air cleaning module **200**. The control module **400** may be located above the first air cleaning module **100** and below the second air cleaning module **200**.

[0170] Because the control module **400** is disposed between the first air cleaning module **100** and

the second air cleaning module **200**, the lengths of wires connecting the control module **400** to the first and second air cleaning modules **100** and **200** and variation in the lengths of the wires may be minimized, and the wiring structure may be simplified.

[0171] Further, because the filter pressing plates **440** and **450** are integrated with the upper and lower covers **410** and **420**, vibration and noise may be further reduced, and manufacturing costs may be reduced.

[0172] When the first filter **120** is mounted to the first filter mounting part **160** of the first air cleaning module **100**, the first filter **120** presses the lower filter pressing plate **450**. When the second filter **220** is mounted to the second filter mounting part **260** of the second air cleaning module **200**, the second filter **220** presses the upper filter pressing plate **440**. Accordingly, the filters **120** and **220** may be more securely supported.

[0173] In addition, the control module **400** includes an upper UVC LED **481** and a lower UVC LED **482**. The upper UVC LED **481** outputs UVC light upwardly toward the second air cleaning module **200** and the second filter **220**. The lower UVC LED **482** outputs UVC light downwardly toward the first air cleaning module **100** and the first filter **120**.

[0174] In general, ultraviolet light may be subdivided according to wavelength. For example, ultraviolet light may be classified into UVA having a wavelength of 320 nm to 400 nm, UVB having a wavelength of 280 nm to 320 nm, and UVC having a wavelength of 200 nm to 280 nm. In particular, UVC, which is short-wavelength ultraviolet light, has characteristics of destroying DNA of bacteria and causing a chemical reaction with a specific substance, and thus is effective for sterilization.

[0175] The control module **400** includes a mounting part **460** that supports the main PCB **430**. Various sensors, such as an LED sensor **471**, a laser sensor **472**, and a gas sensor **473**, may be mounted to the mounting part **460**. Some of the sensors (e.g., the LED sensor **471**) may include a cover **465**.

[0176] The LED sensor **471** may emit infrared LED light and may detect light scattered by dust, thereby sensing the amount of dust. The laser sensor **472** may sense the amount of dust using laser light. The gas sensor **473** may sense the amount of gas in the air.

[0177] Although the LED sensor **471**, the laser sensor **472**, and the gas sensor **473** are illustrated in FIGS. **14** and **15**, the disclosure is not limited thereto. The types of sensors provided and combinations thereof may be varied. In addition, one or more communication modules (not shown), such as a Wi-Fi module, may be provided inside the control module **400**.

[0178] FIG. **16** is an assembled view of the second air cleaning module according to the embodiment of the present disclosure, and FIGS. **17** and **18** are exploded views of the second air cleaning module according to the embodiment of the present disclosure.

[0179] Referring to FIGS. **16** to **18**, the second discharge port **210a** that is open upward is formed in an upper portion of the second air cleaning module **200**. The upper discharge grille **210** that changes the direction of air flowing upward is disposed inside the second discharge port **210a**.

[0180] For example, the second discharge port **210a** may be formed in a ring shape. The upper discharge grille **210** may be formed in a radial shape. The hollow top cover **270** is disposed so as to surround the upper discharge grille **210**. The booster module **300** may be coupled to the center of the upper discharge grille **210**. The head rotating part **380** may be disposed in the central cavity of the upper discharge grille **210** and may be coupled to the booster neck **360** supporting the booster module **300**.

[0181] Air with foreign matter removed therefrom by the second filter **220** is discharged upwardly through the second discharge port **210a**.

[0182] In the second air cleaning module **200**, the second filter **220** is mounted to the second filter mounting part **260**. The above description of the first filter **120** and the first filter mounting part **160** may be similarly applied to the second filter **220** and the second filter mounting part **260**.

However, while the first filter **120** and the first filter mounting part **160** are located above the first

blower fan **140** and the first fan housing **145**, the second filter **220** and the second filter mounting part **260** are located below the second blower fan **240** and the second fan housing **245**.

[0183] The second air cleaning module **200** includes a second fan housing **245** disposed above the second filter **220**, a second blower fan **240** rotatably disposed in the second fan housing **245**, and a second fan motor **230** configured to rotate the second blower fan **240**.

[0184] In addition, the second air cleaning module **200** may further include a wire cover **215** that protects a wire connected to the second fan motor **230**. The second air cleaning module **200** may further include a rotary bearing **290** that reduces frictional force generated during rotation.

[0185] The second blower fan **240** suctions air in the axial direction and discharges the air upwardly in the radial direction. The second fan housing **245** may form a fan path that accommodates the second blower fan **240** and guides the air moved by the second blower fan **240** upwardly.

[0186] The second fan motor **230** may be supported by the second motor fastening part **225**. The rotation shaft of the second fan motor **230** may extend downward from the second fan motor **230** and may pass through the top of the second motor fastening part **225** to be connected to the second blower fan **240**.

[0187] Second magnets **280** are disposed on two opposite sides of the second fan housing **245**. The second magnets **280** may be coupled to metallic pieces of the first case **11**. In addition, the second magnets **280** may be coupled to metallic pieces of the second case **12**.

[0188] In some embodiments, each of the second magnets **280** may include a plurality of magnets. Some of the plurality of magnets may be coupled to the metallic pieces of the first case **11**, and the remaining ones of the plurality of magnets may be coupled to the metallic pieces of the second case **12**. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the second fan housing **245**, and the magnets may be disposed at the first case **11** and the second case **12**.

[0189] A plurality of support portions **250** extends downward from the second fan housing **245**. For example, the support portions **250** may be beams elongated in the longitudinal direction. A wire cover **255** may be disposed on at least one surface of each of the plurality of support portions **250** so as to cover the support portions **250**. The support portions **250** and the wire cover **255** may be spaced a predetermined interval from each other to define a space in which wires or the like are disposed.

[0190] A steel net **265** is disposed between the second filter **220** and the second blower fan **240** in order to prevent body parts of the user or any other object from entering the space in which the second blower fan **240** is disposed.

[0191] FIGS. **19** to **21** are views for explaining discharge of purified air through the bottom of the air cleaner according to the embodiment of the present disclosure.

[0192] FIG. **19** is a view showing the flow path of the lower discharge airflow, and FIG. **20** is a view visualizing the lower discharge airflow using a laser. FIG. **21** is an enlarged view of a lower discharge airflow guide structure.

[0193] Referring to FIGS. **19** to **21**, the first discharge port **110a** is formed in the lower end of the first air cleaning module **100**, and the lower discharge grille **110** is disposed inside the first discharge port **110a** in order to guide the lower discharge airflow.

[0194] The lower discharge grille **110** may include a plurality of grilles bent at a predetermined angle in the outward direction. The lower discharge grille **110** may be disposed inside the first discharge port **110a** in order to deliver the lower discharge airflow in the lateral direction.

[0195] When the first blower fan **140** is rotated by the first motor **130**, the air purified by the first filter **120** is discharged through the first discharge port **110a** as the lower discharge airflow.

[0196] The airflow guide **170a** is disposed on the upper surface of the bottom plate **170** in order to guide air discharged downwardly through the first discharge port **110a** in the lateral direction.

[0197] General allergen particles, which have a size of about 3 μm to about 100 μm , are larger and heavier than indoor floating dust particles, which have a size of about 1 μm , and thus tend to settle on the floor. For example, mold particles have a size of about 10 μm , household dust allergen particles have a size of about 15 μm , and pollen particles have a size of about 50 μm .

[0198] Because virus or cigarette smoke particles have a very small size, the same tend to continuously float without settling on the floor. Indoor floating dust particles remain suspended in the air for a very long time. For example, it takes about 9 to 10 hours for indoor floating dust particles to settle from the ceiling to the floor of the room.

[0199] However, under the same conditions, mold particles, household dust allergen particles, and pollen particles float for a short time of several seconds to several minutes and then settle on the floor. Therefore, it is not easy for the conventional air cleaner to capture allergen particles. Conventionally, the only way to remove the settled particles is to use a vacuum cleaner.

[0200] According to the present disclosure, the lower airflow discharged downwardly from the first air cleaning module **100** may cause allergen particles to float in the air, and the floating allergen particles may be captured by the filters **120** and **220**.

[0201] Meanwhile, according to an embodiment of the present disclosure, the speed of the lower discharge airflow may be controlled in response to the terminal velocity of allergen particles. The speed of the lower discharge airflow may be set to be higher than the terminal velocity of allergen particles.

[0202] The rotational speed of the first blower fan **140** may be controlled based on the terminal velocity of allergen particles. The rotational speed of the first blower fan **140** may correspond to the speed of the lower discharge airflow set to be higher than the terminal velocity of allergen particles. The controller mounted on the main PCB **430** of the control module **400** may control the rotational speed of the first blower fan **140**.

[0203] The terminal velocity of allergen particles may be calculated using balance between gravity, buoyancy, and drag force acting on allergen particles. For example, the terminal velocity of allergen particles may be calculated using Stokes' terminal velocity formula.

[0204] In order to cause mold particles having a size of about 10 μm and household dust allergen particles having a size of about 15 μm to float in the air, the speed of the lower discharge airflow may be set to be higher than the terminal velocity of larger household dust allergen particles. For example, the speed of the lower discharge airflow may be set to 0.0038 m/s or higher.

[0205] The airflow guide **170a** includes a curved portion **173**, a flat portion **171** extending laterally from the curved portion **173**, and a boss **172** protruding upward from an end portion of the flat portion **171**.

[0206] The boss **172** may form an airflow that effectively causes allergen particles to float in the air. The boss **172** may reduce flow resistance at the end portion of the flat portion **171**.

[0207] The boss **172** may prevent flow separation at the end portion of the flat portion **171**. The boss **172** may be formed in a ring shape along the edge of the airflow guide **170a**. The boss **172** may protrude from the flat portion **171** by a predetermined height **d1**, thereby preventing the occurrence of turbulence at the end portion of the flat portion **171** and causing the airflow to ascend. Accordingly, it is possible to effectively cause allergen particles settled on the floor to float in the air while preventing flow separation.

[0208] If the height **d1** of the boss **172** is too great, it may excessively increase the degree of ascending of the airflow, which is inappropriate for causing allergen particles settled on the floor to float in the air. Therefore, it is desirable that the height **d1** of the boss **172** be set to several millimeters (mm) or less, so long as the boss **172** is capable of preventing flow separation.

[0209] The curved portion **173** may be formed so as to surround the central pillar **175** of the bottom plate **170**. The curved portion **173**, which imparts a curvature to the airflow guide **170a**, may be formed so as to be round with a predetermined radius of curvature **R1**. The radius of curvature **R1** may be set so as to guide the lower discharge airflow to smoothly flow in the lateral direction.

[0210] The grille bending angle of the lower discharge grille **110** may be set to 20 degrees to 40 degrees so that the airflow rises at 20 degrees or greater in order to cause allergen particles on the floor to float in the air.

[0211] The radius of curvature **R1** may be set corresponding to the grille bending angle of the lower discharge grille **110**. For example, a predetermined angle **ag1** based on the vertical direction may be determined corresponding to the grille bending angle of the lower discharge grille **110**, and the radius of curvature **R1** may be set so as to implement the determined angle **ag1**. In more detail, the angle **ag1** may be set to be smaller than the grille bending angle of the lower discharge grille **110**.

[0212] The predetermined angle **ag1** may be an angle between the central pillar **175** and the curved portion **173**. In this case, the predetermined angle **ag1** may be smaller than the angle of the discharge airflow through the discharge port **110**. The angle of the discharge airflow may correspond to the grille arrangement angle and grille bending angle of the lower discharge grille **110**.

[0213] The curved portion **173** may guide the lower discharge airflow to smoothly flow in the lateral direction, thereby reducing flow resistance. According to the present disclosure, in order to deliver the lower discharge airflow in the lateral direction, the bottom plate **170** is imparted with a curvature, and the boss **172** is formed at an end portion of the flat portion **171**. Accordingly, a turbulent flow for causing allergen particles to float in the air may be created. That is, because the airflow guide **170a** is imparted with an inclination and a curvature and is provided with the boss for preventing flow separation, resistance to the lower discharge airflow may be minimized.

[0214] FIGS. **22A** to **22E** are views for explaining performance comparison between a case in which purified air is discharged through the upper and lower ends of the product and a case in which purified air is discharged only through the upper end of the product.

[0215] In FIGS. **22A** to **22E**, Case **1** corresponds to a case in which purified air is discharged through the upper and lower ends of the product, and the allergen removal performance over time of Case **1** is shown in the left region **2210**. In FIGS. **22A** to **22E**, Case **2** corresponds to a case in which purified air is discharged only through the upper end of the product, and the allergen removal performance over time of Case **2** is shown in the right region **2220**.

[0216] FIG. **22A** shows the initial state, FIG. **22B** shows the state after the elapse of 1 second, FIG. **22C** shows the state after the elapse of 15 seconds, FIG. **22D** shows the state after the elapse of 60 seconds, and FIG. **22E** shows the state after the elapse of 300 seconds.

[0217] Referring to FIGS. **22A** to **22E**, it can be seen that a flow stagnant zone A occurs on the floor after the elapse of a short floating time of allergen particles in the case in which purified air is discharged only through the upper end of the product.

[0218] In contrast, in the case in which purified air is discharged through the upper and lower ends of the product, since air smoothly circulates in the room, it is possible to more effectively remove allergen particles, which are greatly influenced by an airflow.

[0219] When the lower discharge airflow is applied, the performance of removing allergen particles of 2.8 μm or greater is greatly improved (by about 18% or greater), compared to improvement of the performance of removing fine dust (0.3 μm).

[0220] FIGS. **23** and **24** are views for explaining the case coupling structure according to the embodiment of the present disclosure. FIG. **23** shows the magnets and the metallic pieces in the state in which the case is separated therefrom, and FIG. **24** is a cross-sectional view showing the coupling state of the magnets and the metallic pieces when the case coupled thereto is viewed from above.

[0221] The air cleaning modules **100** and **200** include first magnets **180** and second magnets **280**. The second magnets **280** are disposed above the first magnets **180**. The first air cleaning module **100** may include the first magnets **180**, and the second air cleaning module **200** may include the second magnets **280**.

[0222] For example, the first magnets **180** may be disposed on two opposite sides of the first fan housing **145** of the first air cleaning module **100**. In addition, the second magnets **280** may be disposed on two opposite sides of the second fan housing **245** of the second air cleaning module **200**.

[0223] The first magnets **180** may be coupled to the metallic pieces **15** of the first case **11**. In addition, the first magnets **180** may be coupled to the metallic pieces **15** of the second case **12**.

[0224] The second magnets **280** may be coupled to the metallic pieces **15** of the first case **11**. In addition, the second magnets **280** may be coupled to the metallic pieces **15** of the second case **12**.

[0225] In some embodiments, each of the first magnets **180** may include a plurality of magnets. Some of the plurality of magnets may be coupled to the metallic pieces of the first case **11**, and the remaining ones of the plurality of magnets may be coupled to the metallic pieces of the second case **12**. The positions of the magnets and the metallic pieces may be set to be opposite those exemplified above. For example, the metallic pieces may be disposed on two opposite sides of the first fan housing **145**, and the magnets may be disposed at the first case **11** and the second case **12**.

[0226] The first case **11** may include lower metallic pieces **15b** coupled to the first magnets **180** and upper metallic pieces **15a** coupled to the second magnets **280**.

[0227] Similarly, the second case **12** may also include lower metallic pieces **15b** coupled to the first magnets **180** and upper metallic pieces **15a** coupled to the second magnets **280**.

[0228] The lower metallic pieces **15b** of the first case **11** and the second case **12** are coupled to the first magnets **180**. The upper metallic pieces **15a** of the first case **11** and the second case **12** are coupled to the second magnets **280**.

[0229] In the case of the conventional 2-stage air cleaner, each of the air cleaning modules requires individual cases. Therefore, demounting and mounting processes are complicated and time-consuming. For example, a total of eight coupling parts may be included in the cases, which increases the number of components (magnets and metallic pieces). Further, because the intermediate portion of the product appears discontinuous, the aesthetics of the product deteriorates.

[0230] In order to solve this problem, according to the present disclosure, the upper/lower outer grille is implemented in an integral form in each of the air cleaning modules, rather than being divided into two parts, whereby the aesthetics of the product is improved, the number of coupling components is reduced, and use convenience is improved. According to the present disclosure, each of the first case **11**, which is the front case, and the second case **12**, which is the rear case, is implemented in an integral form as a common cover for the air cleaning modules **100** and **200**, whereby demounting and mounting processes may be simplified, and manufacturing costs may be reduced. Further, according to the present disclosure, it is possible to check the internal state of the product by separating only one of the first case **11** and the second case **12**.

[0231] The control module **400** may be disposed between the first air cleaning module **100** and the second air cleaning module **200**. The control module **400** may include a dust sensor **474**. For example, the dust sensor **474** may include at least one of the LED sensor **471**, the laser sensor **472**, or the gas sensor **473** mentioned above.

[0232] According to the present disclosure, some of the support parts **150** and **250** may be formed in a truss structure in which a plurality of linear members is arranged in a plurality of triangular forms, thereby securing sufficient mechanical rigidity and increasing the suction area.

[0233] Hereinafter, the structure of the support parts will be described in detail with reference to the drawings.

[0234] FIGS. **25** to **28** are views for explaining the structure of the support parts according to the embodiment of the present disclosure.

[0235] Referring to FIGS. **25** to **28**, the air cleaning modules **100** and **200** include fan housings **145** and **245** and a plurality of first support parts **2510** extending from the fan housings **145** and **245** in the longitudinal direction.

[0236] One side of each of the first support parts **2510** may be connected to the fan housings **145** and **245**, and the other side of each of the first support parts **2510** may be connected to the control module **400**. Accordingly, the first support parts **2510** may support the filters **120** and **220** mounted to the filter mounting parts **160** and **260** from the lateral direction.

[0237] The first support parts **2510** are formed in a truss structure in which a plurality of linear members is arranged in a plurality of triangular forms. At least two first support parts **2510** may be disposed on the side surfaces of the filters **120** and **220** so as to face each other.

[0238] The air cleaning modules **100** and **200** may include one or more second support parts **2520** extending from the fan housings **145** and **245** in the longitudinal direction and a wire cover **2530** covering at least one surface of each of the second support parts **2520**. The wire cover **2530** may be disposed so as to be spaced a predetermined interval from one side of each of the second support parts **2520**.

[0239] In order to secure sufficient space for accommodating wires or the like, the width sb2 of the second support part **2520** may be larger than the width sb1 of the first support part **2510**. Because at least one surface of the second support part **2520** is shielded by the wire cover **2530**, the second support part **2520** may be implemented as a general beam column, rather than being formed in a truss structure.

[0240] Each of the first air cleaning module **100** and the second air cleaning module **200** may include the first support part **2510**.

[0241] The first support part **2510** of the first air cleaning module **100** may extend upward from the first fan housing **145**. The first support part **2510** of the second air cleaning module **200** may extend downward from the second fan housing **245**.

[0242] If a general column structure is applied to the side surfaces of the filters **120** and **220**, the suction areas of the filters **120** and **220** may decrease in inverse proportion to the area of the column, and the air cleaning performance may be reduced. However, according to the present disclosure, since the first support part **2510** is formed in a truss structure, additional suction areas may be secured in the side surfaces of the filters **120** and **220**, and flow resistance caused by shielding of the column may be minimized.

[0243] As is apparent from the above description, according to at least one of the embodiments of the present disclosure, an air cleaner includes a first discharge port formed in the bottom thereof and a second discharge port formed in the top thereof, and air with foreign matter removed therefrom is discharged downwardly through the first discharge port formed in the bottom and is discharged upwardly through the second discharge port formed in the top. Accordingly, an air return phenomenon may be prevented.

[0244] According to at least one of the embodiments of the present disclosure, a first air cleaning module configured to discharge purified air through the bottom thereof is disposed at a lower position, and a second air cleaning module configured to discharge purified air through the top thereof is disposed at an upper position. Accordingly, it is possible to prevent air discharged from the first air cleaning module from being introduced into the second air cleaning module and to improve air cleaning performance.

[0245] According to at least one of the embodiments of the present disclosure, because a booster module including a fan is disposed on the top of the air cleaning module, it is possible to send purified air farther away.

[0246] According to at least one of the embodiments of the present disclosure, because an airflow guide is disposed on the upper surface of a bottom plate, which is disposed below the air cleaning module so as to be in contact with the floor, in order to guide air discharged downwardly through the first discharge port in the lateral direction, allergens causing allergies may be caused to float in the air and captured. Accordingly, it is possible to effectively remove allergens.

[0247] According to at least one of the embodiments of the present disclosure, because the airflow guide includes a boss protruding upward from an end portion thereof, it is possible to reduce flow

resistance.

[0248] According to at least one of the embodiments of the present disclosure, because a lower discharge grille including a plurality of grilles bent in the outward direction is disposed inside the first discharge port, it is possible to create a lower discharge airflow for floating of allergen particles.

[0249] According to at least one of the embodiments of the present disclosure, because the booster module is configured such that a display panel is disposed so as to cover the entirety of the front surface of a booster fan and air is discharged through a slit formed beside the display panel, it is possible to prevent dust from accumulating on the booster fan and the inner grille.

[0250] According to at least one of the embodiments of the present disclosure, because the booster module is removably mounted on the top of the second air cleaning module, cleaning and maintenance of the product may be facilitated.

[0251] According to at least one of the embodiments of the present disclosure, because the booster module further includes a power module that comes into contact with a terminal of the second air cleaning module when mounted and receives power from the second air cleaning module, demounting and mounting of the booster module may be further facilitated.

[0252] According to at least one of the embodiments of the present disclosure, because a case is commonly used to cover the outer circumferential surfaces of the air cleaning modules stacked vertically, rather than being divided into parts for the respective air cleaning modules, coupling and separation of the case may be facilitated, and manufacturing costs may be reduced.

[0253] According to at least one of the embodiments of the present disclosure, because the case is conveniently coupled and separated using a small number of magnets and metallic pieces, maintenance and repair of internal components may be facilitated.

[0254] According to at least one of the embodiments of the present disclosure, because some of a plurality of support parts extending vertically from the fan housing are formed in a truss structure, the air suction area of the filter may be increased.

[0255] The effects achievable through the disclosure are not limited to the above-mentioned effects, and other effects not mentioned herein will be clearly understood by those skilled in the art from the description of the claims.

[0256] Although the present disclosure has been described with reference to specific embodiments shown in the drawings, it is apparent to those skilled in the art that the present disclosure is not limited to those exemplary embodiments and is embodied in many forms without departing from the scope of the present disclosure, which is described in the following claims. These modifications should not be individually understood from the technical spirit or scope of the present disclosure.

Claims

1. An air cleaner comprising: a case including a suction port disposed on a peripheral surface of the case; a first air cleaning module disposed in the case, the first air cleaning module comprising a first blower fan, a first filter configured to remove foreign matter from air introduced through the suction port, and a first discharge port disposed at a bottom of the first air cleaning module, the first discharge port being configured to discharge air filtered by the first filter in a downward direction; and a second air cleaning module disposed in the case, the second air cleaning module comprising a second blower fan, a second filter configured to remove foreign matter from air introduced through the suction port, and a second discharge port disposed at a top of the second air cleaning module, the second discharge port being configured to discharge air filtered by the second filter in an upward direction, and the second air cleaning module being disposed above the first air cleaning module.

2. The air cleaner according to claim 1, further comprising a bottom plate disposed below the first air cleaning module and configured to be in contact with a floor, the bottom plate comprising an

airflow guide disposed on an upper surface of the bottom plate, the airflow guide being configured to guide air discharged downwardly through the first discharge port in a lateral direction.

3. The air cleaner according to claim 2, wherein the airflow guide comprises: a curved portion; a flat portion extending laterally from the curved portion; and a boss protruding upward from an end portion of the flat portion.

4. The air cleaner according to claim 1, further comprising a lower discharge grille disposed inside the first discharge port, the lower discharge grille comprising a plurality of grilles bent in an outward direction.

5. The air cleaner according to claim 4, wherein the plurality of grilles are arranged in a concentric circle structure.

6. The air cleaner according to claim 1, further comprising a booster module disposed above the second air cleaning module, the booster module being configured to control a direction of air discharged through the second discharge port.

7. The air cleaner according to claim 6, wherein the booster module comprises: an outer cover; an inner grille disposed inside the outer cover, the inner grille comprising a discharge port; a rear suction grille comprising a suction port; a booster fan disposed at a front of the rear suction grille and disposed inside the inner grille; and a motor configured to rotate the booster fan.

8. The air cleaner according to claim 7, wherein the booster module further comprises: a display panel disposed at a front of the booster fan; and an inner flow guide disposed between the outer cover and the inner grille, the inner flow guide being configured to change a flow direction of air discharged through the discharge port of the inner grille, wherein the discharge port of the inner grille is configured to discharge air to pass through a slit disposed between the inner flow guide and the display panel.

9. The air cleaner according to claim 8, wherein the display panel covers an entire front surface of the booster fan.

10. The air cleaner according to claim 9, wherein the motor is disposed inside the booster fan.

11. An air cleaner comprising: a case including a suction port disposed on a peripheral surface of the case; an air cleaning module disposed in the case, the air cleaning module comprising: a blower fan; a filter configured to remove foreign matter from air introduced through the suction port; a first discharge port disposed in a bottom of the air cleaning module, the first discharge port being configured to discharge air in a downward direction; and a second discharge port disposed in a top of the air cleaning module, the second discharge port being configured to discharge air in an upward direction; and a booster module disposed above the air cleaning module, the booster module being configured to receive at least a portion of the air discharged by the second discharge port.

12. The air cleaner according to claim 11, further comprising a bottom plate disposed below the air cleaning module and configured to be in contact with a floor, the bottom plate comprising an airflow guide disposed on an upper surface of the bottom plate, the air flow guide being configured to guide air discharged downwardly through the first discharge port in a lateral direction.

13. The air cleaner according to claim 12, wherein the airflow guide comprises: a curved portion; a flat portion extending laterally from the curved portion; and a boss protruding upward from an end portion of the flat portion.

14. The air cleaner according to claim 11, further comprising a lower discharge grille disposed inside the first discharge port, the lower discharge grille comprising a plurality of grilles bent in an outward direction.

15. The air cleaner according to claim 14, wherein the plurality of grilles are arranged in a concentric circle structure.

16. The air cleaner according to claim 11, wherein the booster module comprises: an outer cover; an inner grille disposed inside the outer cover, the inner grille comprising a discharge port; a rear suction grille comprising a suction port; a booster fan disposed at a front of the rear suction grille

and disposed inside the inner grille; and a motor configured to rotate the booster fan.

17. The air cleaner according to claim 16, wherein the booster module further comprises: a display panel disposed at a front of the booster fan; and an inner flow guide disposed between the outer cover and the inner grille, the inner flow guide being configured to change a flow direction of air discharged through the discharge port of the inner grille, wherein the air discharged through the discharge port of the inner grille is configured to pass through a slit formed between the inner flow guide and the display panel.

18. The air cleaner according to claim 17, wherein the display panel covers an entire front surface of the booster fan.

19. The air cleaner according to claim 16, wherein the motor is disposed inside the booster fan.

20. The air cleaner according to claim 11, wherein the booster module is removably mounted to the top of the air cleaning module, the booster module further comprising a power module configured to come into contact with a terminal of the air cleaning module and receive power from the air cleaning module.
